Shared Tucker Factors for Tucker Tensor Trains: Algorithms, Bounds, and Proofs

Companion notes for the T3Toolbox sharing implementation

November 13, 2026

Abstract

These notes summarize the mathematics behind adding *shared Tucker factors* to the Tucker tensor train (T3) format: the shared rank theorem, the grouped truncation algorithm and its quasi-optimality bound, initialization from unshared representations, the geometry of the fixed-rank shared manifold (tied tangent spaces, the Gram-sum projection, retraction, dimension), the edge-conditioning statistics used for rank continuation, and the single-pass property of group-aware useless-rank removal. The shared format is the SF-ETT decomposition of Molozhavenko and Rakhuba [1], generalized from one trailing shared block to an arbitrary partition of the modes into sharing groups.

Revision (v2, 2026-08-19): updated after implementation review — Proposition 5.2 restated basis-free (no complement basis), the corewise geometry added (Proposition 5.7), the numerical-form and zero-padded-restart remarks added (Remarks 5.5 and 6.4), the dimension formula's limits corrected (Proposition 5.11 and its erratum remark), the tied doubled-rank embedding made explicit (Remark 5.10), and the tail-energy definition of $\mathrm{rank}_\varepsilon$ *made explicit (Theorem 3.4).*

1 Setup and notation

A *Tucker tensor train* (T3; equivalently, extended tensor train) represents a tensor $\boldsymbol{T} \in \mathbb{R}^{N_1 \times \cdots \times N_d}$ as

$$\boldsymbol{T} = \big[\!\big[G;\ B_1, \ldots, B_d \big]\!\big], \qquad G = \mathrm{TT}(G_1, \ldots, G_d),$$

where $B_i \in \mathbb{R}^{N_i \times n_i}$ are the Tucker factors, $G_i \in \mathbb{R}^{r_{i-1} \times n_i \times r_i}$ are the TT cores of the central Tucker core $G \in \mathbb{R}^{n_1 \times \cdots \times n_d}$, and $r_0 = r_d = 1$. We call $\boldsymbol{n} = (n_1, \ldots, n_d)$ the Tucker ranks and $\boldsymbol{r} = (r_1, \ldots, r_{d-1})$ the TT ranks.

$\boldsymbol{T}_{(i)} \in \mathbb{R}^{N_i \times \prod_{j \neq i} N_j}$ denotes the i-th matricization of $\boldsymbol{T}$ and $\boldsymbol{T}^{\langle i \rangle}$ its i-th matrix unfolding. $W_i \in \mathbb{R}^{n_i \times r_{i-1} r_i}$ denotes the *up-matricization* of core G_i (Tucker leg to rows). All statements below assume the standard *orthogonalized representations*: the factors B_i have orthonormal columns, and, when a statement refers to mode i, the TT chain is in *i-orthogonal form* (cores left of i left-orthogonal, cores right of i right-orthogonal), reachable by QR sweeps without changing $\boldsymbol{T}$. In the tangent-space statements of Section 5 the factors and their directions are written D_i, D_g, $\dot{D}$: the down store of T3Toolbox's orthonormal frame (T4S [4], App. A.3). At a point in orthonormal representation D_i is B_i; the corewise geometry of Proposition 5.7, whose points are the raw cores, keeps B_i.

T4S writes U, O, P, Q for our D, U, L, R; its δU is our V (`tucker_variations`), its δG our H (`tt_variations`). Before T3Toolbox 2026.5.0 this note wrote U for the factors (now B,

and D in the tangent statements), B_i for the block K_i of Algorithm 4.1, and B for the tied coordinates Z.

Definition 1.1 (Mode environment / Gram matrix)**.** $\Gamma_i := G_{(i)}G_{(i)}^\top \in \mathbb{R}^{n_i \times n_i}$, the Gram of the i-th matricization of the Tucker core. Γ_i is representation-independent; in the i-orthogonal representation it telescopes to

$$\Gamma_i = W_i W_i^\top.$$

We write S_i for any factor with $\Gamma_i = S_i S_i^\top$ (e.g. the R-factor absorbed during mode-i down-orthogonalization; in frame coordinates of the fixed-rank manifold, S_i is pre-absorbed so that the coordinate Gram is the identity).

Definition 1.2 (Sharing partition; shared T3)**.** A *sharing partition* assigns each mode $i \in \{1, \dots, d\}$ to a group g; modes in a group $g = (i_1, \dots, i_{k_g})$ must have equal mode sizes, denoted N_g. A T3 is *shared* (an SF-T3) if $B_i = B_g$ and $n_i = n_g$ for all $i \in g$, for every group. Singleton groups impose no constraint. $\mathcal{M}^{\mathrm{sh}}_{\boldsymbol{n},\boldsymbol{r}}$ denotes the set of shared T3s with fixed ranks (Section 5).

Throughout, "the concatenation" for a group g means the column-wise block matrix $[\,\boldsymbol{T}_{(i_1)} \,|\, \cdots \,|\, \boldsymbol{T}_{(i_{k_g})}\,]$.

2 Shared ranks

Lemma 2.1 (Stacked skeleton factorization)**.** *Let $\boldsymbol{T}$ be a shared T3 with orthonormal factors, and let each W_i be taken in the i-orthogonal representation. Then for each mode i of group g,*

$$\boldsymbol{T}_{(i)} = B_g\, W_i\, M_i, \tag{1}$$

where M_i has orthonormal rows, and stacking the group's matricizations,

$$\big[\,\boldsymbol{T}_{(i_1)} \,|\, \cdots \,|\, \boldsymbol{T}_{(i_k)}\,\big] = B_g\,\big[\,W_{i_1} \,|\, \cdots \,|\, W_{i_k}\,\big]\ \mathrm{blockdiag}\big(M_{i_1}, \dots, M_{i_k}\big). \tag{2}$$

Consequently the nonzero singular values of the concatenation equal those of the small matrix $[\,W_{i_1}|\cdots|W_{i_k}\,]$, and

$$s_{g,j}^2 \;=\; \lambda_j\Big(\textstyle\sum_{i\in g}\Gamma_i\Big) \;=\; \lambda_j\Big(\textstyle\sum_{i\in g} S_i S_i^\top\Big), \qquad j = 1, \dots, n_g, \tag{3}$$

where $s_{g,1} \geq s_{g,2} \geq \cdots$ are the singular values of the concatenation.

Proof. Equation (1) is the standard matricization identity for a T3 in i-orthogonal form: the environment of mode i's Tucker leg contracts to a matrix M_i with orthonormal rows (a Kronecker/interface product of orthonormal-column factors and left/right-orthogonal partial chains). Each block of the concatenation is factored in its own representation of the same tensor $\boldsymbol{T}$; since each factorization holds individually and the left factor B_g is common by sharedness, (2) follows by placing the M_i in block-diagonal position. The outer factors in (2) are, respectively, orthonormal-column and orthonormal-row, so singular values pass to the middle factor. Finally, $[\,W_{i_1}|\cdots|W_{i_k}\,][\,W_{i_1}|\cdots|W_{i_k}\,]^\top = \sum_{i\in g} W_i W_i^\top = \sum_{i\in g}\Gamma_i$, and Γ_i is representation-independent, so mixing per-mode representations is consistent. □

Theorem 2.2 (Shared minimal rank; cf. [1, Thm. 1], [2])**.** *The minimal Tucker rank of group g over all shared T3 representations of $\boldsymbol{T}$ is*

$$n_g^{\min} \;=\; \mathrm{rank}\big[\,\boldsymbol{T}_{(i_1)} \,|\, \cdots \,|\, \boldsymbol{T}_{(i_k)}\,\big].$$

TT ranks are unaffected by sharing: $r_i^{\min} = \mathrm{rank}\,\boldsymbol{T}^{\langle i\rangle}$.

Proof sketch. ($\geq$): in any shared representation, every $\boldsymbol{T}_{(i)}$, $i \in g$, has column space contained in range(B_g), hence so does the concatenation, giving rank$[\cdots] \leq n_g$. ($\leq$): take B_g to be an orthonormal basis of the concatenation's column space; projecting each mode ($B_g B_g^\top \boldsymbol{T}_{(i)} = \boldsymbol{T}_{(i)}$) yields an exact shared representation with $n_g = \text{rank}[\cdots]$; TT-decompose the resulting core at the unfolding ranks. See [1, 2] for details. □

3 Grouped truncation (shared rounding)

Algorithm 3.1 (Grouped T3-SVD / SF-rounding; cf. [1, Alg. 1])**.** Input: shared T3 with orthonormal factors; target ranks. Output: shared T3 with target ranks.

(1) TT-round the cores to the target TT ranks (standard sweep; factors untouched). In an interlaced implementation, steps (2)–(4) are woven into the same sweep; the interlacing order changes truncated results but not the bounds below.

(2) For each mode i, obtain W_i in the i-orthogonal representation (one orthogonalization sweep).

(3) Singleton modes: truncated SVD of W_i; rotate factor and core as usual.

(4) Each group g: one truncated SVD of the concatenation $[\,W_{i_1}|\cdots|W_{i_k}\,] \in \mathbb{R}^{n_g \times \sum_{i\in g} r_{i-1}r_i}$, yielding leading left singular vectors $Y \in \mathbb{R}^{n_g \times n'_g}$. Update $B_g \leftarrow B_g Y$ *once* (assigned to every mode of g) and apply $Y^\top$ to the Tucker leg of every core in g.

By Lemma 2.1, step (4) computes the optimal rank-n'_g truncation of the concatenation while touching the large dimension N_g only in the single product $B_g Y$, at cost $O(N_g n_g^2)$ per group; the SVD itself acts on an $n_g \times \sum_i r_{i-1}r_i$ matrix.

Theorem 3.2 (Quasi-optimality; [1, Thm. 2])**.** *Let $P_{\text{sf}}(\boldsymbol{T})$ denote the output of Algorithm 3.1 (equivalently, of its dense analogue applied to an arbitrary tensor) at ranks $(\boldsymbol{n}, \boldsymbol{r})$. Then*

$$\big\|\boldsymbol{T} - P_{\text{sf}}(\boldsymbol{T})\big\|_{\text{HS}} \;\leq\; C(d) \inf_{\substack{\boldsymbol{Y}\ \text{shared},\\ \text{ranks}(\boldsymbol{Y})\preceq(\boldsymbol{n},\boldsymbol{r})}} \big\|\boldsymbol{T} - \boldsymbol{Y}\big\|_{\text{HS}}, \qquad C(d) = \sqrt{d} + \sqrt{d}\sqrt{d-1} + \sqrt{d-1}.$$

Proof sketch. Split $P_{\text{sf}} = P_{\text{SF-Tucker}} \circ P_{\text{TT}}$ (either order; [1, Rem. 1]). Each stage is quasi-optimal in its own class ($\sqrt{d}$ for the HOSVD-type shared factor truncation, using that the shared-mode error is controlled by the concatenation tail; $\sqrt{d-1}$ for TT-SVD), and each stage's best-in-class error is bounded by the best shared error via the triangle inequality; the constants compose to $C(d)$. □

Remark 3.3 (Truncation error bound). When the input is exactly shared and step (4) truncates group g from n_g to n'_g with retained basis Y, the squared error contributed by that step is *at most* $\sum_{j>n'_g} s_{g,j}^2$: by Lemma 2.1 the discarded tail of the concatenation equals the *sum over the group's modes* of the single-mode projection errors, $\sum_{i\in g} \|(I - YY^\top)\boldsymbol{T}_{(i)}\|_F^2$, which bounds the error of the simultaneous projection $\boldsymbol{T} \times_{i\in g} YY^\top$ by the standard HOSVD subadditivity argument. Equality holds for singleton groups ($k = 1$); for $k > 1$ the simultaneous projection error is strictly smaller in general (verified numerically: e.g. $4.19 \cdot 10^2$ vs the tail bound $4.80 \cdot 10^2$ on a random tied example).

Theorem 3.4 (Rank upper bound for the interlaced sweep)**.** *In tolerance-based truncation, every rank selected by Algorithm 3.1 satisfies*

$$n'_g \;\le\; \operatorname{rank}_\varepsilon\big[\,\boldsymbol{T}_{(i_1)}\,|\,\cdots\,|\,\boldsymbol{T}_{(i_k)}\,\big] \quad \textit{(of the original input)}, \qquad r'_i \;\le\; \operatorname{rank}_\varepsilon \boldsymbol{T}^{\langle i\rangle},$$

where $\operatorname{rank}_\varepsilon$ *is numerical rank at the working tolerance, counted by tail Frobenius energy (the criterion the implemented truncated SVD uses):* $\operatorname{rank}_\varepsilon(A) = \min\{\, r : \|A - A_r\|_F \le \max(\varepsilon\|A\|_F,\ \mathrm{atol})\,\}$ *with* A_r *the best rank-*r *approximation.*

Proof sketch. Edge monotonicity: any truncation performed at another edge multiplies the group's concatenation on the right by a block-orthogonal-projection-type factor (and on the left by nothing, since factors are orthonormal), which cannot increase its rank; likewise for unfoldings. Hence the matrix whose SVD is truncated at each step has rank at most that of the corresponding matricization/unfolding of the original tensor. □

4 Initialization: unshared → shared

Algorithm 4.1 (Shared projection of an unshared T3)**.** Input: T3 with orthonormal but distinct factors B_i, $i \in g$. For each group g:

(1) Stack factors and compute one tall QR: $[\,B_{i_1}|\cdots|B_{i_k}\,] = QR$, with $Q \in \mathbb{R}^{N_g \times m}$, $m \le \sum_{i\in g} n_i$.

(2) Express each block $K_i := B_i W_i$ (never formed at size N_g) in the Q basis: $Q^\top K_i = R_{[i]} W_i$, where $R_{[i]}$ is the corresponding column block of R. SVD the small stacked matrix $[\,R_{[i_1]}W_{i_1}|\cdots|R_{[i_k]}W_{i_k}\,] \in \mathbb{R}^{m \times \sum_i r_{i-1} r_i}$; retain leading left singular vectors Y.

(3) Set $B_g := QY$; project each mode by applying $(B_g^\top B_i)$ to the Tucker leg of core i and replacing the factor by B_g. Follow with Algorithm 3.1.

Lemma 4.2. *The column space of* $[\,K_{i_1}|\cdots|K_{i_k}\,]$ *lies in* $\operatorname{range}(Q)$*, so step (2) computes the exact dominant left singular subspace of the concatenation* $[\,\boldsymbol{T}_{(i_1)}|\cdots|\boldsymbol{T}_{(i_k)}\,]$ *while touching the large dimension only in the QR of step (1) and the product* QY*. The resulting projection is quasi-optimal with respect to the best shared approximation, with the constant of Theorem 3.2 (same composition argument, with the shared-HOSVD stage now acting on unshared input).*

Remark 4.3 (Implementation form: exact rewrite, defer all selection to the rounding)*.* T3Toolbox implements only step (1)'s stacked-factor SVD, as an *exact* common-span rewrite: the factor coefficients are the SVD's own row blocks ($B_i = W_{[i]}\operatorname{diag}(s)\,V^\top$, valid for arbitrary — not necessarily orthonormal — input factors), the shared factor becomes $V^\top$ (group rank $m = \min(\sum_i n_i, N_g)$), and ALL rank/tolerance selection is deferred to the grouped rounding (Algorithm 3.1). This is equivalent to steps (2)–(3): the optimal shared basis lies in the span of the group's factors, and the group spectrum is representation-independent, so the rounding on the rewritten representation selects the same subspace and reports the same s_g; the explicit small SVD of steps (2)–(3) is an optional efficiency device (pre-shrinking the group rank before the TT sweep), at the cost of a second truncation stage.

5 Fixed-rank geometry with tied factors

Let $\mathcal{M}_{\boldsymbol{n},\boldsymbol{r}}$ denote the manifold of fixed-rank T3s and $T_{\boldsymbol{T}}\mathcal{M}_{\boldsymbol{n},\boldsymbol{r}}$ its tangent space at a point $\boldsymbol{T}$ with orthonormal representation. A tangent vector is a sum of d TT-variation terms and d

Tucker-variation terms; in *gauged* form the Tucker variation at mode i satisfies $D_i^\top \dot{V}_i = 0$, and gauge conditions make all cross terms in the Hilbert–Schmidt inner product vanish, so

$$\langle \xi, \eta \rangle_{\mathrm{HS}} = \sum_{i=1}^{d} \mathrm{tr}\big(\dot{V}_i^\top \dot{W}_i \, \Gamma_i\big) + (\text{TT-variation terms}). \tag{4}$$

In *frame coordinates* the factor S_i (with $\Gamma_i = S_i S_i^\top$) is pre-absorbed per mode, so the coordinates C_i of Tucker variations are orthonormal: corewise Euclidean inner products equal Hilbert–Schmidt inner products.

Definition 5.1 (Shared manifold)**.** $\mathcal{M}^{\mathrm{sh}}_{\boldsymbol{n},\boldsymbol{r}} \subset \mathcal{M}_{\boldsymbol{n},\boldsymbol{r}}$ is the set of shared T3-representable tensors with fixed shared ranks: one Tucker rank n_g and one factor per group. It is the transversal intersection of the fixed-rank T3 manifold with the shared-Tucker manifold and is smooth [1, Thm. 5].

Proposition 5.2 (Tied tangent space is a tilted subspace)**.** *Fix a shared point. A tangent vector is tangent to $\mathcal{M}^{\mathrm{sh}}_{\boldsymbol{n},\boldsymbol{r}}$ iff its TT variations are unrestricted and, per group, its Tucker variations arise from a single* gauged ambient direction $\dot{D} \in \mathbb{R}^{N_g \times n_g}$ *with $D_g^\top \dot{D} = 0$: in the (S-absorbed) frame coordinates,*

$$C_i = \dot{D}\, S_i \quad \text{for a common gauged } \dot{D}, \qquad i \in g. \tag{5}$$

This is a linear subspace of the full gauged tangent space, tilted mode-by-mode by the (generally distinct, generally rectangular) S_i.

Remark 5.3 (No complement basis; rectangular S_i). An earlier draft stated (5) through an orthonormal complement O_g of D_g ($\dot{D} = O_g Z$, coordinates $C_i = Z S_i$), suggesting a complement must be constructed once per group and tied across it. The subspace depends only on D_g (tied by hypothesis) and the S_i; no complement basis is ever materialized (nor is one materialized anywhere in T3Toolbox), and the per-mode rotational freedom in the factorization $\Gamma_i = S_i S_i^\top$ (replacing S_i by $S_i Q_i$ together with the correspondingly rotated orthonormal core) cancels identically in (5) and in the projection (6). Note also that in implementation $S_i \in \mathbb{R}^{n_g \times n_{U,i}}$ with $n_{U,i} = \min(n_g,\ r_{i-1} r_i)$ — rectangular in general, so an individual Gram Γ_i may be singular at a perfectly smooth shared point; only the group sum is invertible there (Proposition 6.1(iii)).

Proposition 5.4 (Projection onto the tied subspace)**.** *The orthogonal projection (in the frame-coordinate Euclidean metric, which equals Hilbert–Schmidt) of arbitrary gauged coordinates $(C_i)_{i \in g}$ onto the subspace (5) is given by*

$$\dot{D} \;=\; \Big(\sum_{i \in g} C_i S_i^\top\Big)\Big(\sum_{i \in g} S_i S_i^\top\Big)^{-1} \;=\; \Big(\sum_{i \in g} C_i S_i^\top\Big)\Big(\sum_{i \in g} \Gamma_i\Big)^{-1}, \qquad C_i \leftarrow \dot{D} S_i. \tag{6}$$

TT variations and singleton modes are unaffected, and $\dot{D}$ is automatically gauged when the C_i are. Formula (6) is the projection formula of [1, §5.3] rederived as least squares onto a tilted diagonal; $\sum_i C_i S_i^\top$ equals the sum of raw per-mode cotangents $(\nabla f)_{(i)} G_{(i)}^\top$.

Proof. Minimize $\sum_{i \in g} \|C_i - \dot{D} S_i\|_F^2$ over $\dot{D}$; setting the gradient to zero gives the normal equations $\sum_i C_i S_i^\top = \dot{D} \sum_i S_i S_i^\top$, invertible whenever the point has full shared rank ($s_{g,n_g} > 0$ by (3)). Gauge: $D_g^\top \sum_i C_i S_i^\top = \sum_i (D_g^\top C_i) S_i^\top = 0$, and the Gram sum acts on the other side, so $D_g^\top \dot{D} = 0$. □

Remark 5.5 (Numerical form: solve the tilted least squares by SVD, never by normal equations). Implementations should not form the Gram sum. Stacking

$$M_g = \begin{bmatrix} S_{i_1}^\top \\ \vdots \\ S_{i_k}^\top \end{bmatrix} \in \mathbb{R}^{(\sum_{i\in g} n_{U,i})\times n_g}, \qquad C^{\text{stack}} = \begin{bmatrix} C_{i_1}^\top \\ \vdots \\ C_{i_k}^\top \end{bmatrix},$$

the projection is the least-squares solution $\dot{D}^\top = M_g^+ C^{\text{stack}}$ (then $C_i^\top \leftarrow S_i^\top \dot{D}^\top$, the corresponding row block of $M_g \dot{D}^\top$), computed from one thin SVD of M_g with a relative singular-value clip. Three payoffs. (i) The solve has the intrinsic least-squares sensitivity (κ_g on the solution component); the normal-equations/Cholesky route is κ_g^2 unconditionally. (ii) $\sigma(M_g) = (s_{g,j})_j$ *is* the group spectrum (3), at full (non-squared) accuracy: in float32 the Gram route loses trailing levels below $\sqrt{\varepsilon}\, s_{g,1}$ entirely (measured at $s_{g,\min}/s_{g,1} \approx 10^{-4}$: relative error $3 \cdot 10^{-1}$ via the Gram eigendecomposition vs $8 \cdot 10^{-5}$ via the SVD of M_g). (iii) The clipped pseudoinverse is well-defined (minimum-norm) at rank-deficient points, which rank continuation visits by construction (Remark 6.4).

Remark 5.6 (Metric unchanged). Because (5) is a linear subspace and the frame metric is Euclidean on gauged coordinates, the restriction of the metric to tied tangents requires no correction: inner products and norms of tied tangents are computed exactly as in the unshared case. The Gram-sum enters only through the projection (6). In particular, plain averaging $C_i \mapsto \frac{1}{k}\sum_j C_j$ projects onto the wrong subspace $\{C_{i_1} = \cdots = C_{i_k}\}$, which corresponds to no tied ambient direction.

Proposition 5.7 (The corewise geometry ties by averaging). *Consider instead the over-parametrized* corewise *geometry: points are the raw cores $((B_i)_i, (G_i)_i)$, the metric is Euclidean on the core entries, and the retraction is addition of core perturbations. The shared parameter set $\{\, B_i = B_j,\ i,j \in g,\ \forall g \,\}$ is a linear subspace of the parameter space; its tangent space is $\{\dot{B}_i$ equal within each group$\}$ (TT-core perturbations unrestricted), and the orthogonal projection onto it in the corewise metric is the per-group arithmetic mean $\dot{B}_i \mapsto \frac{1}{k_g}\sum_{j\in g} \dot{B}_j$. The additive retraction then preserves tying exactly.*

Remark 5.8 (One principle, two formulas). Both geometries tie by *orthogonal projection onto the geometry's tied subspace in the geometry's own metric on its own coordinates.* The formulas differ because the coordinates differ: the manifold geometry's gauged coordinates carry S_i preabsorbed, so its tied subspace is tilted (5) and the projection is the Gram-weighted solve (6); the corewise coordinates are raw factor copies, so its tied subspace is the diagonal and the projection is the mean. Using either formula in the other geometry projects onto the wrong subspace; applied to the same gauged coordinates, the arithmetic mean matches the Gram-weighted solve only when the S_i themselves coincide across the group (equal Grams Γ_i are not sufficient – the S_i are pinned to the per-mode orthonormal cores and generally differ by more than a common rotation even at exactly symmetric points).

Proposition 5.9 (Retraction). *For a shared point $\boldsymbol{T}$ and tied tangent ξ, the attached point $\boldsymbol{T}+\xi$ admits a shared T3 representation with doubled ranks (the doubled factor is $[\, D_g \,|\, \dot{D}_g \,]$, identical across each group), and*

$$R(\boldsymbol{T}, \xi) \coloneqq P_{\text{sf}}(\boldsymbol{T} + \xi)$$

(Algorithm 3.1 truncating back to the original ranks) is a retraction on $\mathcal{M}^{\text{sh}}_{\boldsymbol{n},\boldsymbol{r}}$. This follows from quasi-optimality (Theorem 3.2) via the argument of [3, Prop. 2.3]; see [1, Prop. 6].

Remark 5.10 (Tied doubled-rank embedding). The attached point $\boldsymbol{T}+\xi$ should be *constructed* tied, not re-tied after the fact: per group, the doubled factor is $[\,D_g \,|\, \dot{D}\,]$ (one matrix, used at every mode of g; $2n_g$ columns) and the paired core block at mode i is the *center core* W_i (equivalently, S_i absorbed back into the S-stripped orthonormal core) — cf. the SF-ETT tangent representation [1, §5.2]. The naive embedding by per-mode factor blocks $[\,D_g \,|\, \dot{D}S_i\,]$ (the S-absorbed coordinates) is *not* tied: the blocks differ across the group in value and, when the $n_{U,i}$ differ, even in shape, and no per-mode averaging repairs this. $\dot{D}$ is recovered from tied coordinates by the least-squares solve of Remark 5.5 (exact on tied input; its residual is a natural tied-tangent check).

Proposition 5.11 (Dimension; cf. [1, Thm. 5]). *With groups g of mode size N_g and shared rank n_g (singleton groups reducing to the standard terms),*

$$\dim \mathcal{M}^{\mathrm{sh}}_{\boldsymbol{n},\boldsymbol{r}} = \underbrace{\sum_{i=1}^{d} r_{i-1}\, n_{g(i)}\, r_i \;-\; \sum_{i=1}^{d-1} r_i^2}_{\text{TT-core term (cores are never tied)}} \;+\; \sum_{\text{groups } g} n_g\big(N_g - n_g\big),$$

where $g(i)$ is the group of mode i and the boundary bonds $r_0 = r_d = 1$ carry no gauge: one Stiefel-quotient term per group *instead of per mode.*

Remark 5.12 (Erratum and scope). The printed [1, Thm. 5] subtracts $\sum_{i=1}^{d} r_i^2$, over-subtracting by exactly 1 (the boundary bond $r_d = 1$ is not a gauge degree of freedom); the matrix case $d = 2$ pins the limits above. Note also that both [1] and [2] prove smoothness and the dimension for a *single trailing block* of shared modes; the arbitrary-partition statement here follows by iterating the same transversality argument group by group, and has been validated empirically by matching the rank of a dense tied tangent basis to the formula — including a case where a group rank exceeds an individual mode's local ceiling $r_{i-1}r_i$ (Proposition 7.1), where the *unshared* minimal-rank reduction would clip the rank and the formula, applied to the unshared-reduced ranks, under-counts. Implementations must therefore reduce by the *shared* minimal ranks before applying the formula.

6 Rank continuation: edge conditioning under sharing

The rank-continuation policy of [4] grows ranks on the most well-conditioned edges of the tensor network, keeping all edges comparably conditioned, because edge conditioning governs the conditioning of the fixed-rank subproblem. Sharing replaces the group's per-mode Tucker edges with a single edge whose spectrum is the concatenation spectrum $s_{g,\cdot}$ of (3):

$$\kappa_g \coloneqq \frac{s_{g,1}}{s_{g,n_g}}, \qquad \kappa_i^{\mathrm{loc}} \coloneqq \sqrt{\mathrm{cond}(\Gamma_i)} \quad (i \in g),$$

competing in the same growth rule as TT-edge and singleton-Tucker condition numbers: an edge grows only if it is well conditioned relative to the worst edge, $\kappa < \kappa_{\max}/\tau$ (with $\tau > 1$); the group contributes *one* κ_g to the pool, and one growth decision is applied group-wide.

Proposition 6.1 (κ_g is the shared subproblem conditioning). *At a shared point:*

(i) The Gram of the tied parameterization $Z \mapsto (ZS_i)_{i\in g}$ is $\sum_{i\in g}\Gamma_i$: for a tied Tucker tangent, $\|\xi\|_{\mathrm{HS}}^2 = \mathrm{tr}\big(Z(\sum_i \Gamma_i)Z^\top\big)$. Its eigenvalue spread is κ_g^2.

(ii) The projection solve (6) has condition number κ_g^2.

(iii) $s_{g,n_g} \to 0$ iff the point approaches the stratum of lower shared rank (Theorem 2.2), where the fixed-rank parametrization degenerates. Individual Γ_i are never inverted for shared modes; per-mode near-rank-deficiency is not a degeneracy of $\mathcal{M}^{\mathrm{sh}}_{\boldsymbol{n},\boldsymbol{r}}$.

Proof. (i) is (4) with $\dot{V}_i \leftrightarrow ZS_i$ absorbed into frame coordinates: $\sum_i \|ZS_i\|_F^2 = \mathrm{tr}(Z(\sum_i S_i S_i^\top)Z^\top)$. (ii) is immediate from (6) and (3). (iii) is Theorem 2.2 with (3). □

Proposition 6.2 (Conditioning of the sum)**.** *$\kappa_g \le \max_{i\in g} \kappa_i^{\mathrm{loc}}$, and the inequality can be strict by an arbitrarily large factor.*

Proof. All Γ_i are positive semidefinite, so $\lambda_{\max}(\sum_i \Gamma_i) \le \sum_i \lambda_{\max}(\Gamma_i)$ and $\lambda_{\min}(\sum_i \Gamma_i) \ge \sum_i \lambda_{\min}(\Gamma_i)$. Hence, by the mediant inequality $\frac{\sum_i a_i}{\sum_i b_i} \le \max_i \frac{a_i}{b_i}$ (for $a_i \ge 0$, $b_i > 0$),

$$\kappa_g^2 = \frac{\lambda_{\max}(\sum_i \Gamma_i)}{\lambda_{\min}(\sum_i \Gamma_i)} \le \frac{\sum_i \lambda_{\max}(\Gamma_i)}{\sum_i \lambda_{\min}(\Gamma_i)} \le \max_{i\in g} \frac{\lambda_{\max}(\Gamma_i)}{\lambda_{\min}(\Gamma_i)} = \max_{i\in g} \mathrm{cond}(\Gamma_i).$$

Strictness: $\Gamma_1 = \mathrm{diag}(1,\epsilon)$, $\Gamma_2 = \mathrm{diag}(\epsilon,1)$ gives $\kappa_1^{\mathrm{loc}} = \kappa_2^{\mathrm{loc}} = \epsilon^{-1/2}$ but $\kappa_g = 1$. □

Proposition 6.2 says the shared edge is never worse conditioned than the worst per-mode Tucker edge of its group, and can be far better: under tying, a direction is well-determined if *some* mode of the group informs it. Conversely, κ_g remaining small while n_g grows past each mode's individual effective rank signals a shared basis being forced to cover a union of dissimilar column spaces (sharing a poor model for the data).

Corollary 6.3 (Scale cancellation; exact symmetric degeneration)**.** *Each i-orthogonal representation preserves norms, so $\sum_j s_{g,j}^2 = k_g \|\boldsymbol{T}\|_{\mathrm{HS}}^2$: raw magnitudes inflate by $\sqrt{k_g}$ relative to a single mode but cancel in the ratio κ_g, which compares against TT and singleton edges with no normalization. If the group is exactly symmetric ($\Gamma_{i_1} = \cdots = \Gamma_{i_k} = \Gamma$; e.g. symmetric derivative tensors with all input modes shared), then $\sum_i \Gamma_i = k\,\Gamma$, so $s_{g,j} = \sqrt{k}\,\sigma_j$ elementwise and κ_g equals the per-mode condition number exactly.*

Remark 6.4 (Zero-padded restarts: which tangent channels are alive). Rank continuation warm-starts by zero-padding, which places the iterate *on* the lower-shared-rank stratum: the freshly added factor directions carry no core mass, so the corresponding rows of every S_i vanish, the new levels of s_g are exactly zero, and the tied Tucker channel (5) admits no first-order motion in the new directions (the clipped pseudoinverse of Remark 5.5 returns the minimum-norm solution — zero there). The escape is carried by the *untied TT-variation channel*: TT variations are unrestricted and have first-order content in the new Tucker slots, paired with the (deterministically completed, tied) new factor columns, so the first Gauss–Newton step activates the new directions through core mass; once s_g acquires mass, the tied Tucker channel reactivates, with a transiently large κ_g that the clipped solve handles (small mass ⇒ large permitted rotation — the same behavior the unshared S-absorbed coordinates exhibit). Consequently, full shared rank must *not* be enforced as a precondition of the projection, and the projection solve must be a clipped pseudoinverse. All statements in this remark have been verified numerically on zero-padded shared points.

7 Group-aware useless-rank removal: single pass suffices

Rank proposals are reduced to nondegenerate values by the standard three-phase sweep (Tucker vs. mode size; TT right-unfoldings right-to-left; interleaved Tucker/TT left-to-right), with local ceilings $n_i \le \min(N_i,\, r_{i-1}r_i)$ (singletons), $r_i \le r_{i-1}n_i$, $r_{i-1} \le n_i r_i$.

Proposition 7.1 (Group ceiling)**.** *For a group g,*

$$n_g \;\le\; \min\Big(N_g,\; \sum_{i\in g} \min\big(N_g,\; r_{i-1}\, r_i\big)\Big). \tag{7}$$

Proof. By Theorem 2.2 and (1), $\operatorname{rank}[\,\boldsymbol{T}_{(i_1)}|\cdots|\boldsymbol{T}_{(i_k)}\,] \le \min(N_g,\, \sum_i \operatorname{rank}\boldsymbol{T}_{(i)}) \le \min(N_g,\, \sum_i \min(N_g,\, r_{i-1}r_i))$. □

A shared basis column is useless only if useless for *every* mode of the group: the per-mode ceilings *add* across the group rather than min-reduce, and n_g may exceed $r_{i-1}r_i$ at individual modes without the representation being reducible. Ceiling (7) couples n_g to TT ranks at distant chain positions, so the constraint incidence graph acquires cycles; generic monotone constraint propagation on cyclic graphs requires fixed-point iteration, in contrast to the tree-structured unshared case (dynamic programming on a tree; one ordered pass). Nevertheless:

Lemma 7.2 (Self-protection)**.** *After the cap $n_g \leftarrow \min\big(N_g,\; \sum_{i\in g}\min(N_g,\, r_{i-1}r_i)\big)$, every TT constraint at the group's own modes involving n_g holds automatically, given that all n_g values used in the TT phases satisfied $n_g \le N_g$ (phase 1).*

Proof. The sum dominates each of its own terms, so $n_g \ge \min(N_g,\, r_{i-1}r_i)$ for each $i \in g$. *Case $r_{i-1}r_i \le N_g$:* then $n_g \ge r_{i-1}r_i$, and since ranks are ≥ 1,

$$r_{i-1}\, n_g \;\ge\; r_{i-1}^2\, r_i \;\ge\; r_i, \qquad n_g\, r_i \;\ge\; r_{i-1}\, r_i^2 \;\ge\; r_{i-1},$$

so both the left- and right-unfolding constraints hold. *Case $r_{i-1}r_i > N_g$:* the i-th term equals N_g, hence the sum is $\ge N_g$ and $n_g = N_g$ exactly; the constraints $r_i \le r_{i-1}N_g$ and $r_{i-1} \le N_g\, r_i$ are then precisely those enforced by phase 1 ($n \le N$) together with the standard TT phases, since every n_g value used there was $\le N_g$. □

Theorem 7.3 (Single pass)**.** *The three-phase sweep with (a) group-wide Tucker reductions, (b) the ceiling (7), and (c) re-evaluation and re-capping of n_g at every group-mode visit during the left-to-right phase (standard Tucker-then-TT ordering at each core), reaches the componentwise-greatest nondegenerate rank vector below the proposal in a single pass.*

Proof. Monotone cap iteration from the proposal converges to the greatest fixed point (Knaster–Tarski/Kleene on the finite lattice of integer rank vectors); it remains to show one ordered pass attains it, i.e. that no constraint is violated at exit. Constraints not involving any n_g are handled exactly as in the unshared tree argument. For constraints involving n_g:

(i) *Staleness within the sweep.* n_g is monotone nonincreasing through the sweep and $\le N_g$ from phase 1 onward. A TT cap at group mode i was enforced with the then-current value; the left-to-right phase touches r_i for the last time at position i, before any later re-evaluation can reduce n_g; final consistency at mode i then follows from Lemma 7.2 applied with final values, since $n_g^{\text{final}} \ge \min(N_g,\, r_{i-1}r_i)$ with the final (frozen) r_{i-1}, r_i.

(ii) *Last group mode.* The ceiling at the group's last mode i_k is evaluated before that mode's own TT cap; if the cap binds ($r_{i_k} \leftarrow r_{i_k-1} n_g$), the mode's ceiling term becomes $\min(N_g, r_{i_k-1}^2\, n_g) \geq \min(N_g, n_g) = n_g$, so the just-set n_g remains consistent.

(iii) *No propagation.* n_g appears in no constraint outside group g's modes, and by Lemma 7.2 ceiling-driven reductions of n_g never force TT reductions. Hence the direction $r \to n_g$ flows while $n_g \to r$ is vacuous; every cycle of the constraint incidence graph contains a vacuous edge and is never traversed, and no cross-group cascade exists.

At exit, all constraints hold, so the pass output is a fixed point; since each cap is the loosest reduction consistent with its constraint and caps are monotone, the output is the greatest fixed point below the proposal. □

Remark 7.4. The proof uses the specific phase structure: phase 1 ($n \leq N$) *precedes* the TT phases, and the left-to-right phase applies the Tucker step before the TT step at each core. Implementations should assert the theorem empirically (a second pass changes nothing) rather than assume it survives reorderings. Note also that sharing only ever *relaxes* Tucker ceilings relative to per-mode removal, since the sum in (7) dominates each per-mode term.

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